Aura: The Empathic AI Companion

Template: (**4.1: Orchestrating AI Models to Achieve a Goal)**

Introduction

Recent advances in large language models have enabled conversational agents that are increasingly fluent and context-aware. However, most deployed systems remain primarily **text-based** or **task-oriented**, offering limited emotional awareness and minimal support for natural, human-like interaction. At the same time, growing interest in AI companionship, emotionally aware customer support, and socially assistive technologies highlights the need for conversational systems that can engage users not only cognitively, but also **emotionally**.

This project explores the design and implementation of a **general-purpose emotionally intelligent chatbot** capable of real-time, multimodal interaction through **speech, facial expression, and audiovisual feedback**. While the system draws inspiration from AI companion paradigms, it is intentionally designed to remain **domain-agnostic**, supporting a wide range of interaction contexts—from emotionally sensitive scenarios such as eldercare or social practice, to reliability-focused applications such as customer support. Rather than embedding domain-specific behaviors into the model itself, the system enables flexible adaptation through configurable prompts and modular components.

The motivation for this work is both personal and practical. Existing assistants such as Siri or Alexa excel at functional tasks but lack emotional presence, while current AI companions often rely on text-only or audio-only interfaces that limit conversational realism. This project investigates whether **adding a visual, emotionally expressive interface**—combined with affect-aware dialogue generation—can make human–AI interaction feel more natural, engaging, and responsive. At a broader level, increasing levels of loneliness and social anxiety further motivate the exploration of emotionally aware conversational systems that can support safe and accessible interaction.

The project follows the template ***Orchestrating AI Models to Achieve a Goal,*** focusing on the integration of multiple pretrained AI components rather than developing new models from scratch. Speech recognition, emotion recognition, large language model reasoning, and audiovisual synthesis are combined into a single real-time pipeline, with particular emphasis on latency, modularity, and practical deployability. Through this orchestration-based approach, the project aims to demonstrate how emotionally aware multimodal interaction can be achieved under realistic computational constraints.

Literature Review

Motivation and Background

Artificial intelligence has rapidly evolved from utilitarian virtual assistants into emotionally responsive conversational partners. With advances in large language models (LLMs), chatbots now engage users in increasingly natural and socially attuned dialogue—ranging from everyday informational queries to sustained, affectively rich companionship. Unlike early voice assistants such as Alexa or Siri, contemporary LLM-based companions can adapt tone, emotional expressiveness, and conversational framing, enabling interactions that approximate human–human emotional exchange. Although LLMs do not experience genuine emotion, fine-tuned models can emulate empathic responses across diverse relational contexts, from friendships and mentorship to more intimate or unconventional forms of companionship.

This shift raises critical questions about whether emotionally aware AI systems can help alleviate forms of social anxiety and loneliness. Loneliness is traditionally defined as a subjective state arising from the discrepancy between desired and actual social connection (Perlman & Peplau, 1982). In principle, any interaction—human or artificial—that meets an individual's need for social presence may temporarily relieve loneliness. While AI should not replace genuine human relationships, it may serve as a supportive tool for individuals experiencing elevated social anxiety, difficulty forming connections, or transitional periods of isolation. Empirical evidence reinforces this possibility: recent work demonstrates that even short conversations with an LLM-based companion can significantly reduce perceived loneliness—**producing effects statistically comparable to talking with another person** (De Freitas et al., 2025). Importantly, these effects persisted across multiple days of repeated interaction, suggesting that users formed durable socio-emotional engagement rather than responding to fleeting novelty. As generative AI technologies continue to integrate into smartphones and immersive digital environments, their potential to offer scalable, personalized emotional support outside clinical settings becomes increasingly relevant.

This potential societal value of such companionship is especially pronounced among older adults. Loneliness and social isolation increase substantially with age due to declining mobility, shrinking social networks, and reduced engagement opportunities (Dykstra et al., 2005). At the same time, many care facilities face limited emotional bandwidth, with human caregivers unable to provide consistent social interaction. These challenges are amplified by broader demographic trends: population aging and declining birth rates are straining care infrastructures in many countries, reducing the ratio of working-age adults available to support a growing elderly population. In this context, AI-mediated companionship offers a complementary source of emotional support that can scale where human resources cannot. Furthermore, psychological research shows that communication preferences among older adults are shaped by individual

personality traits (Balmaceda et al., 2014), and recent findings indicate that they attribute distinct characteristics to conversational agents' voices despite identical acoustic features (Shahid et al., 2025). These insights suggest that customizable voices and personas might allow AI companions to provide personalized support at a scale that human caregivers cannot, enhancing comfort, trust, and long-term engagement.

Beyond personal well-being, emotionally aware AI also has emerging relevance in customer-facing environments as well. Research in marketing and consumer psychology shows that "embodied" chatbot interfaces—agents that communicate with socially rich cues—can foster more favorable consumer–brand relationships and higher satisfaction (Bergner, Hildebrand, & Häubl, 2023). However, this domain also illustrates the risks of misaligned emotional communication. Some studies find that relying on bots can elicit negative consumer reactions due to perceptions that automation prioritizes firm efficiency over customer benefit (Castelo et al., 2023). Others highlight that chatbots may fail when confronted with unanticipated or affectively complex user messages (De Freitas et al., 2023a). These limitations underscore a general challenge: emotionally responsive chat bots must not only simulate empathy but also maintain reliability, contextual sensitivity, and alignment with user expectations.

Across these domains—mental well-being, elderly care, and customer interaction—the literature suggests both a substantial need and a compelling opportunity. Loneliness affects an estimated 30–60% of adults in the United States (Holt-Lunstad, Robles, & Sbarra, 2017; Ipsos, 2021), and similar trends are observed globally. As AI becomes woven into daily life, emotionally capable digital companions could provide accessible, scalable, and user-specific support, supplementing human social structures rather than replacing them. Yet achieving this requires advances in multimodal perception, emotional expressiveness, and alignment with human social cues. These gaps motivate the development of a more emotionally aware digital companion capable of interacting through both video and audio, with the aim of enhancing users' sense of connection while respecting the psychological and ethical boundaries of human–AI relationships.

Review of Existing Works

Emotionally intelligent AI companions are not limited to eldercare or mental-wellbeing support; their capacity for empathetic, context-aware dialogue enables applications across a wide range of domains, including immersive role-playing, customer care, and social engagement. This section reviews three representative lines of work, highlighting their contributions, limitations, and implications for emotionally aware multimodal companions. (see Appendix 1 for summary)

Voice-Based Chatbots for Elderly Companionship

A significant example is the system developed by **Alessa & Al-Khalifa (2023)**, who designed a ChatGPT-powered companion for older adults. The chatbot incorporates strategies such as gamified interaction, proactive notifications, and prompts encouraging self-compassion to support mental healthcare. Their approach resembles services such as *HappyTalks*, which initiate scheduled phone conversations with older adults. Alessa & Al-Khalifa rely on caregivers to complete an onboarding questionnaire, which is then used to craft personalized prompts for the chatbot. This approach is reasonable within a research setting, but scaling it beyond that would require careful safeguards around privacy, data governance, and the ethical handling of sensitive information especially among highly vulnerable groups.

The authors used a voice-first interface because many older adults face significant usability barriers with smartphone keyboards, resulting in frequent input errors. Interaction modes that minimize fine motor requirements—such as voice or video-driven communication—are generally more suitable for elderly users than text-based interfaces.

Their implementation relies on ASR, ChatGPT for dialogue, and a TTS system, though the specific TTS model is not disclosed. While this omission is common in early design-focused studies, it is worth noting that different TTS pipelines offer varying degrees of emotional expressiveness. Some contemporary systems allow controlled rendering of nuanced cues—such as soft sighs, gentle laughter, or subtle shifts in intonation—whereas other models produce a more neutral speaking style. Consequently, depending on the TTS technology selected, the emotional richness of the companion's voice could vary, potentially shaping how users perceive warmth, empathy, or engagement during interaction.

Human evaluators rated the system positively for fluency, coherence, and avoidance of repetition, but identified weaker performance in engagement, inquisitiveness, and sustained conversational interest. The authors also emphasize safety concerns, noting that ChatGPT may struggle with complex emotional or situational queries. Fine-tuning a model on domain-specific emotional datasets would likely improve its reliability when engaging with elderly users.

Emotional Social Chatbots

Historical interest in conversational companions dates to ELIZA (Wang, 2024), yet modern systems like Replika go far beyond scripted responses by enabling highly anthropomorphic interactions. Empirical studies (Pentina, Hancock & Xie 2023; Xie, Pentina & Hancock 2023) show that many users form strong emotional bonds with Replika—sometimes to the point of dependency—particularly when they perceive the chatbot as humanlike. Emotional bonding can offer comfort—similar to how pets help humans cope with grief, loneliness, or fear—but unlike pets, AI systems can unintentionally reinforce dependency or simulate attachment in ways

society and regulatory frameworks are unprepared to govern. Replika therefore represents the **over-attachment extreme**: emotionally rich, yet potentially unhealthy in certain patterns of use.

Character.AI operates at the opposite end of the spectrum: it excels in role-playing and persona-based interactions, often guided by user-generated character descriptions. A large-scale analysis by **Lee & Joseph (2025)** found that many publicly accessible Character.AI bots reproduce problematic tropes. For example, bot personas grouped into “boyfriend” or “husband” categories often exhibit behaviors such as jealousy, infidelity, or aggression in their greeting messages. The study highlights concerns related to gender norms, implicit biases, and even depictions of gender-based violence.

Importantly, the authors do not claim these character descriptions represent user outcomes directly; instead, they analyze the cultural biases encoded in character prompts, many of which target fandoms and narratives appealing to minors. **Regardless, risks such as excessive emotional attachment and amplification of pre-existing online toxicity can be present in AI companions and should be acknowledged and addressed;** for instance, Replika has shown instances of emotional attachment, while Character.AI has been observed amplifying online toxicity.

The proposed system should therefore seek a middle path: emotionally supportive, but bounded; expressive, but safe; personalized, but not addictive.

Empathic Fine-Tuned Dialogue Systems

A third line of work focuses on explicitly modelling empathy in text-based conversational agents. Casas et al. (2021) propose a method that fine-tunes language models on empathic dialogues while incorporating emotion-specific signals during response generation. Despite operating solely in text, their system produces responses rated by human evaluators as more empathetic than baseline models—and in many cases, even more empathetic than human-generated replies. This outcome highlights AI’s potential to act as an attentive and empathetic listener in perceived empathy.

However, the authors outline several persistent challenges: balancing emotional content with informational accuracy, sustaining emotional coherence across multi-turn dialogue, and modelling users’ affective states over entire conversations. Their method also attempts to infer emotion solely from text, which limits accuracy given that human affect is expressed through vocal tone, hesitation, rhythm, facial expression, and other cues. Given these limitations, incorporating multimodal inputs—such as voice analysis or visual expression recognition—could substantially improve emotional attunement. The fine-tuning methodology they present is also directly relevant to this project’s goal of building personalized, emotionally sensitive companions.

Related Technology & Methods

Emotionally aware AI companions require an integration of multiple technologies spanning speech, vision, natural language understanding, expressive generation, and real-time inference. While prior sections highlighted social, psychological, and design motivations for such systems, this section reviews the technical foundations that enable a multimodal companion capable of empathic audio–visual interaction.

Speech Processing: Automatic Speech Recognition (ASR)

Speech-to-text (STT) systems occupy the input layer of the system's pipeline, mediating between user and the core dialogue modules by converting speech into text. Modern ASR models—including Whisper, Conformer-CTC, RNN-Transducer, Paraformer, and Zipformer families—now achieve low word-error rates (WER) and robust noise resistance, making them suitable for real-time conversational agents (Gulati et al., 2020).

ASR systems can be divided into offline and online (streaming) approaches, each with distinct implications for real-time conversational inference. Offline models such as Whisper leverage full-utterance context to achieve strong accuracy and multilingual robustness, but introduce higher latency that can hinder interactive responsiveness. Optimized implementations like Faster Whisper reduce this gap through accelerated decoding and segmented inference, enabling near–real-time usage. In contrast, streaming architectures including RNN-Transducer, Conformer-CTC, Paraformer, and Zipformer perform incremental decoding with low latency, making them better suited for turn-taking and barge-in scenarios. While large pretrained models natively support many languages, streaming models typically require explicit multilingual training or adaptation to maintain performance across diverse linguistic inputs.

A crucial challenge for real-time social dialogue is **turn-taking prediction**. Humans interrupt, pause mid-sentence, or switch roles rapidly. ASR alone cannot reliably infer sentence boundaries; instead, systems typically combine **voice-activity detection (VAD)**, prosodic cues (intonation decline, final lengthening), and predictive models trained on conversational corpora (Gravano & Hirschberg, 2011). These signals help determine whether the system should continue listening, respond, or hold back—critical for avoiding awkward interruptions or long silences, particularly in older adults who may pause longer between utterances.

Visual Processing: Voice and Facial Cues

Emotionally aware chatbot must interpret affect beyond words. Research in affective computing demonstrates that **vocal prosody** (pitch, intensity, spectral tilt, rhythm) and **facial action units** provide reliable indicators of emotional valence and arousal (Schuller et al., 2018). Yet most existing chatbots focus on text alone, limiting sensitivity to subtle emotional states.

Two complementary streams of analysis have shown promise for addressing this gap. **Transformer-based Speech Emotion Recognition** extracts prosodic embeddings associated with stress, sadness, or excitement and has been demonstrated to enhance empathetic response quality when fused with large language models (Wu et al., 2024). **Facial Expression Recognition**, grounded in the Facial Action Coding System, enables detection of micro-expressions and multimodal cues such as smiles, brow tension, or hesitation.

Although emotion recognition remains imperfect, even coarse affective signals (e.g., rising agitation or disengagement) substantially improve responsiveness and help prevent emotionally inappropriate replies—a risk previously highlighted in studies of customer-service and role-playing chatbots.

Dialogue Core: Transformer-Based LLM

The language-understanding and response-generation component is the conceptual core of this project. Emotionally supportive chat bot require consistency across turns, emotional coherence, context sensitivity and guardrails to avoid over-attachment or inappropriate therapeutic authority (Casas et al., 2021; Pentina et al., 2023)

Proprietary hosted models (e.g., ChatGPT, Claude) exhibit strong general conversational abilities but offer limited control over task-specific fine-tuning, which is important for sustaining empathic and stylistically consistent dialogue (Casas et al., 2021). In contrast, self-hosted open-weight transformer models—such as LLaMA-family variants, Mistral, and Falcon—allow direct adaptation to application-specific conversational goals. Following prior work, these models can be fine-tuned using empathy-oriented corpora such as the **EmpatheticDialogues** (ED) dataset adopted in the fine-tuning experiments of Casas et al. Parameter-efficient fine-tuning techniques, including LoRA and QLoRA, support memory and compute efficient adaptation of smaller models (3–7B parameters). Quantization will further help satisfy the low-latency constraints of real-time response generation.

Speech Synthesis: TTS with Controllable Paralinguistic

Speech and video synthesis constitutes the final output layer of the system pipeline and plays a critical role in conveying expressive empathy. Conventional text-to-speech (TTS) systems predominantly generate neutral prosody, which limits their effectiveness in emotionally grounded interactions. Achieving emotional presence in synthesized speech therefore requires **explicit control** over expressive cues such as warmth, gentleness, excitement, and calmness. In this project, we investigate two approaches to expressive speech synthesis.

Style-based TTS systems approximate paralinguistic expression by conditioning generation on reference speech or latent style representations, such as Global Style Tokens (Li et al., 2025). These models extract pitch, rhythm, and other paralinguistic features from a reference audio and

transfer them to new content. In addition, mechanisms such as the Transferable Monotonic Aligner (TMA) enable phoneme duration prediction, which is beneficial for audiovisual synchronization (e.g., **TalkingHead**). While this approach is relatively simple and efficient, it offers limited fine-grained control over specific emotional expressions.

End-to-end TTS models, including diffusion-based and autoregressive architectures (e.g., Orpheus-TTS), incorporate explicit emotional tags and context-dependent controls to generate richer expressiveness, such as laughter, sighs, hesitations, and breathiness (Canopy Labs, 2025). These models produce more natural and emotionally nuanced speech; however, their high computational cost and latency constrain their suitability for real-time applications.

Visual Synthesis: Lip-Syncing Systems

Visual synthesis is another critical component of the system which conveys essential social cues to the user. For this, we will explore two approaches to visual generation: **audio-driven(**MuseTalk**)** and **phoneme-driven**(TalkingHead) synthesis.

Audio-driven generation ranges from GAN-based lip-sync models such as **Wav2Lip** (Prajwal et al., 2020) to more expressive head-motion systems like **MuseTalk**(Zhang et al., 2025) and **SadTalke**r (Zhang et al., 2023), as well as multimodal approaches such as **VASA-1** (Xu et al., 2024). These systems vary in terms of realism, latency, and emotional controllability. While these models achieve high visual realism and accurate lip synchronization, they typically incur high latency which limits their suitability for real-time usage. Moreover, these models typically require the full audio to be generated before video synthesis can begin. Although quantization and pruning can reduce model size and inference cost, this requirement to wait for complete audio output remains a bottleneck for latency-sensitive, real-time applications. In addition, these methods raise significant ethical concerns, as they can animate photorealistic depictions of real individuals, increasing the risk of misuse in deepfake generation.

An alternative approach is **Talking-Head**(met4citizen, n.d.), which synchronizes lip and facial movements using phoneme-to-viseme mappings with explicit timing information. Rather than driving animation directly from raw audio waveforms, Talking-Head relies on discrete viseme sequences and their durations to control facial motion. This design allows facial animation to begin as soon as phoneme timing becomes available, without requiring the full audio signal to be generated in advance. As a result, lip synchronization can be produced incrementally with lower perceived latency than audio-driven generation. Although Talking-Head typically relies on 3D lifelike characters rendered frame-by-frame rather than photorealistic video synthesis, this abstraction can be advantageous from an ethical perspective, as it avoids the risk of animating real individuals.

Design and Planing

Proposed System

As highlighted in the literature review, emotionally aware multimodal chatbots have broad application potential, ranging from customer support and eldercare to mental-health companionship and general social interaction. Rather than constraining the system to a single domain, this work aims to design a system that supports flexible, real-time multimodal interaction. The system is intentionally domain-agnostic, allowing users to tailor behaviors, interaction styles, and application-specific logic through system prompts and other configurable mechanisms. The project focuses on two representative domains—eldercare and customer support—because together they capture both the emotional sensitivity and reliability demands essential to the wider class of applications. These domains are used to ground and validate the system requirements, while the underlying architecture remains intentionally generalizable to other use cases.

Real-time audio and video communication are central to this project's goal of moving beyond text-only systems. Although prior work supports voice interaction(Alessa & Al-Khalifa , 2023), video interaction remains largely unexplored. The new system must fill in this gap despite introducing new latency.

Instead of relying on custom hardware like in Alessa & Al-Khalifa works, the proposed system will use browser based technologies to provide a universally accessible interface across devices. This approach aligns with user habits described in prior work(Alessa & Al-Khalifa , 2023), where smartphone-like interaction lowers adoption barriers.

Following findings by Gravano & Hirschberg (2011), the system will use VAD models to detect sentence boundaries and long pauses, enabling dynamic turn-taking between user-led and agent-led conversation which is necessary to prevent awkward long silances during conversation.

Personalization has been shown to improve perceived empathy and emotional engagement (Casas et al., 2021). However, achieving personalization introduces a tension. On one hand, the system must support personalized responses—such as preferred forms of address or contextual references tied to a user's background—to sustain a sense of empathy. On the other hand, personalization mechanisms must avoid per-user model training, which is costly, data-intensive, and difficult to scale. Consequently, the system requires a strategy that enables user-specific adaptation while remaining lightweight and compatible with a shared language model.

Conversational fluidity requires fast response. Research shows that delays exceeding the range 600-1500 ms impair turn-taking and naturalness (Skantze, 2021; Levinson and Torreira, 2015).

The system must have a time-to-first-utterance of 600-1500ms. Achieving this might require aggressive quantization and reduced model complexity, which comes at the cost of the dialogue core's reasoning depth.

System Components

The system can be divided into 4 layers:

1. **Communication** layer (webRTC)
2. **Input** layer (speech-to-text, emotional recognization)
3. **Processing** layer (dialogue core, LLM), and
4. **Output** layer (text-to-speech, animation).

The frontend remains minimal, providing a browser-based call interface with a single "start call" button and a central video window similar to standard video-calling applications.

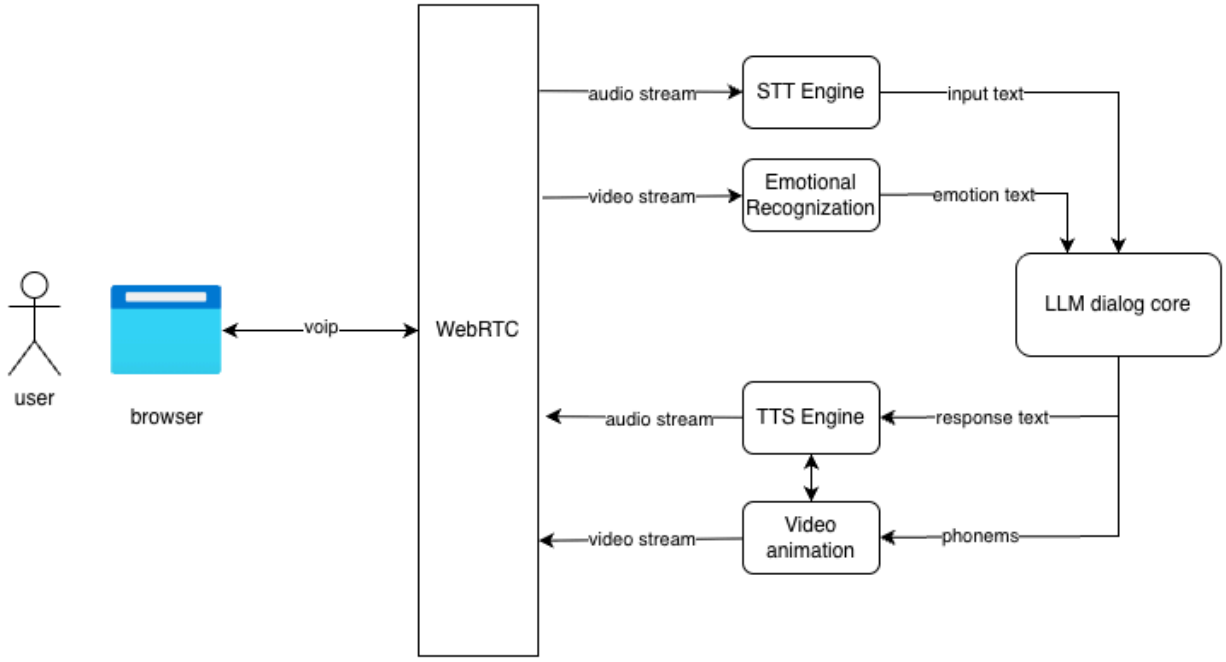


Fig 2.1: Design diagram of Aura

Communication Layer

WebRTC forms the main entry point to the system, handling the transmission and synchronization of audio and video streams over the network using end-to-end protocols and standard codecs. Once the connection is established, the audio and video tracks are forwarded to the subsequent component for transcription, emotion recognition, and response generation.

Input Layer

The input layer consists of a speech-to-text (STT) engine coupled with an emotion recognition module. Audio streams received from the communication layer are processed by the STT system to generate transcriptions while simultaneously extracting emotional nuances from the video signal. Voice activity detection (VAD) is used to identify silences and sentence boundaries; once an utterance is complete, the transcribed sentence and associated affective signals are forwarded to the LLM for response generation.

Among emerging ASR inference engines, **sherpa-onnx**(K2-FSA, n.d.) has gained attention for its flexibility and performance. It supports both streaming and non-streaming inference, as well as multiple model paradigms unifying the decoding stack across different model families. A key motivation for selecting **sherpa-onnx** is its support for online inference. In contrast, the Hugging Face *transformers* library primarily targets offline ASR and requires additional adapter logic for streaming. Moreover, **sherpa-onnx** leverages the ONNX Runtime, which offers efficient CPU-based inference—an important consideration given that GPU resources are primarily reserved for the LLM-driven dialogue engine.

We will explore **Zipformer**, **Paraformer** and **faster-whisper** in this project and evaluate using metrics such as word error rate (WER), language coverage, and end-to-end inference latency on standard datasets such as LibriSpeech and Common Voice.

To extract effective cues from visual input, the system employs a **video-based facial expression recognition module** using the pretrained **DeepFace**(Serengil, n.d.) emotion analysis model. Incoming video frames are processed in a streaming manner, with emotion inference performed on individual frames or at a fixed sampling interval to balance latency and computational cost. The model outputs probability distributions over a set of coarse facial expression categories (e.g., *neutral, happy, sad, angry, fear, surprise, disgust*). To improve robustness under real-time conditions and reduce frame-level noise, temporal smoothing is applied across short frame windows before deriving a dominant affective signal. These visual emotion estimates are treated as **soft contextual cues rather than ground-truth emotional states** and are used to inform downstream dialogue generation without overriding linguistic content or system safeguards.

Processing Layer

After the STT module detects an end-of-sentence, it triggers the dialogue core to generate a response. Inference is performed using **vLLM**(Kwon et al., 2023), a high-throughput, low-latency inference engine. Compared to standard Transformer implementations, **vLLM** supports continuous batching for irregular request patterns, achieves more efficient memory usage via PagedAttention, and delivers higher token throughput under concurrent workloads.

For the dialogue core, we focus on lightweight variants of the LLaMA family, which provide a favourable trade-off between conversational capability, parameter count, and hardware constraints.

Personalization is achieved through a hybrid approach that combines model-level fine-tuning with prompt-level personalization. We fine-tune a **TinyLLaMA** model using LoRA on the **EmpatheticDialogues** dataset, which contains rich affective contexts suitable for modelling supportive dialogue. Fine-grained individual personalization is handled via system prompts encoding lightweight user-specific information, such as preferred forms of address or contextual cues. This avoids the cost and data requirements of per-user fine-tuning while enabling personalized, context-aware interactions.

Model performance will be evaluated using end-to-end latency, time to first token (TTFT), model size, and VRAM usage.

Inference latency and memory footprint can be reduced through smaller models and quantization, although these techniques may degrade reasoning ability. To mitigate this, a dual-model setup can be employed: a lightweight conversational model handles immediate responses, while a more capable reasoning model processes complex queries. During reasoning, brief filler utterances (e.g., “let me think…”) can maintain conversational flow and avoid awkward silences. Latency can be further reduced by exploiting partial transcripts from streaming STT to pre-compute KV caches before full transcription, allowing ASR and language-model computation to overlap. These optimizations are noted as potential improvements but remain out of scope for this project.

Output Layer

The output layer is critical for expressiveness and emotional connection between the user and the system and comprises speech and visual synthesis components. Following the literature review, we explore two approaches: audio-driven and phoneme-driven synthesis.

The audio-driven approach combines **OrpheusTTS** with **MuseTalk**. OrpheusTTS supports controllable expressive tags (e.g., <giggle>, <sighs>) to modulate prosody and employs a Transformer architecture with SNAC audio token decoding. The generated speech is passed to

MuseTalk, together with a reference image, to produce a lip-synced video. This approach delivers highly realistic, emotionally rich speech and enables animating any speaker from a single reference photo. However, it is computationally expensive and introduces high latency, limiting suitability for real-time interaction.

The viseme-driven approach uses a **StyleTTS**-based model (**Kokoro**)(Hexgrad, 2025) together with **TalkingHead**(met4citizen, n.d.). Input text is converted into viseme sequences, which StyleTTS uses to generate speech while also predicting per-viseme durations. These visemes and timings are then passed to TalkingHead to produce accurate lip-synchronised animation of a 3D character. Expressive control is indirect: unlike OrpheusTTS, StyleTTS does not support explicit expression tags, and emotion is instead guided by providing reference speech (e.g., sad or happy tone). This results in significantly lower generation latency but more limited expressiveness and requires predefined 3D character assets rather than a single reference image.

Model selection will be guided by Time-to-First-Byte, support for streaming vs offline generation, and end-to-end latency, memory requirement and expressive quality under realistic usage conditions.

Ethic and Privacy consideration

This project involves a multimodal conversational system that integrates speech recognition, dialogue generation, and lip-synchronized avatar animation. While the system is designed for emotionally supportive interaction, several ethical considerations have been identified and will be addressed in its design.

First, the use of **lip-sync and generative media technologies** raises potential concerns related to deepfake misuse. To mitigate this risk, the system exclusively uses **fully synthetic, cartoon-style avatars** generated by the system and does not incorporate images, videos, or likenesses of real individuals. The resulting animations are intentionally non-photorealistic and clearly identifiable as artificial, reducing the risk of deception, impersonation, or identity misuse.

Second, given the system's goal of empathetic interaction—particularly for users experiencing social anxiety—it is essential to prevent **user over-attachment or misrepresentation of system capabilities**. To address this, the system includes **explicit disclaimers** reminding users that they are interacting with an AI agent rather than a human. This helps maintain appropriate expectations and prevents the system from being perceived as a substitute for professional or human support.

Third, the project involves the collection and storage of **speech transcripts and conversation logs** for safety monitoring, auditing, and system improvement. While this introduces potential **privacy risks**, limited personal data may be necessary to support personalization and interaction

quality. To mitigate these risks, transcripts are stored in **secure, encrypted storage**, with access controlled through session-based encryption keys. Where possible, **PII filtering or redaction mechanisms** are applied before long-term storage, and access is restricted through **token-based authentication**.

Overall, while the project acknowledges broader ethical challenges associated with generative AI and emotionally responsive systems, its constrained scope, synthetic media design, transparency mechanisms, and data protection measures ensure that ethical risks remain limited and manageable within the context of this project.

Evaluation

Evaluation combines quantitative engagement metrics with qualitative user ratings. Metrics include conversational turns per session, time to hang-up, frequency of AI-initiated prompts, and occurrences of long silences or early dropouts, capturing engagement, conversational flow, and the system's ability to sustain interaction.

Participants also provide post-session ratings of responsiveness, emotional appropriateness, clarity, and overall satisfaction. Users are randomly selected to reflect the system's general-purpose scope rather than a single domain. Together, these measures provide a practical end-to-end assessment of system performance and perceived user experience.

Work Plan

The project is structured into three sequential stages, each increasing system functionality and complexity.

Stage 1 develops an audio-only prototype integrating streaming STT, silence detection, LLM-based dialogue generation, and TTS. This stage validates the core speech interaction pipeline, modular component communication, and establishes baseline latency and reliability, resulting in the first working prototype.

Stage 2 introduces video communication and lightweight avatar animation to enable multimodal interaction. The focus is on audio–video synchronization and smooth turn-taking with visual cues. Together, Levels 1 and 2 deliver a functional end-to-end system capable of real-time conversation and form the primary project deliverable. This stage spans four weeks.

Stage 3 focuses on optimization, including LLM fine-tuning for empathetic responses, safety guardrails, personalization, and richer audio–visual output cues. This stage is exploratory due to time constraints and also spans four weeks (see Appendix 2).

The final four weeks are dedicated to end-to-end evaluation, result analysis, and addressing user and system feedback.

Feature Prototype

There are two critical components that require validation in this project. The first is the **dialogue core**, which constitutes the central component of the system. Its feasibility depends on whether a large language model (LLM) can be deployed and executed within the available GPU constraints. The second component is the **video and speech synthesis system**, which, while secondary in priority, represents the most complex part of the overall system, as it is responsible for conveying empathy to the user.

To evaluate the dialogue core, we deployed a pretrained **TinyLLaMA** model on a **Tesla T4 GPU** and measured its performance. The model occupied approximately **2 GB of VRAM** and required **11 seconds** to load. Generating a response of **258 tokens** took **2.57 seconds**, corresponding to a throughput of approximately **100 tokens per second**. In typical conversations, the average response length is around **100 tokens**, resulting in a latency that is slighter higher than the target requirement. However, this is expected to improve by approximately **3–5×** through model quantization and the use of a more powerful GPU (e.g., **A100**), bringing performance within the desired latency constraints. The **time to first token** was measured at **0.5 seconds**, which is acceptable for integration with downstream streaming components if necessary. Qualitatively, the generated responses were excessively long and poorly suited for conversational use, indicating the need for **task-specific fine-tuning** to produce shorter, more dialogue-appropriate outputs (see Appendix 3).

The second component evaluated is the **video and speech synthesis system**. As outlined in the system design, both synthesis approaches were explored. For the **audio-driven synthesis pipeline**, experiments were conducted using a Google Colab instance equipped with a **Tesla T4 GPU (15 GB VRAM, 12 GB system RAM)**. In this pipeline, **Orpheus** was used for speech generation, and the resulting audio was subsequently passed to **MuseTalk** for lip-synchronized video generation.

The perceptual quality of the generated speech and video was high, particularly for speech naturalness. However, the associated **memory usage and latency were substantial**. In the two-stage pipeline, audio generation alone required approximately **6.18 GB of VRAM**, with a model load time of **60 seconds** and a generation time of **51 seconds** for **13 seconds of audio**. This high latency is likely attributable to the use of a **Transformer-based architecture combined with a SNAC decoder**, which is computationally expensive given that it is on a Tesla T4 GPU. Although Canopy Labs reports near–real-time streaming performance (~200 ms latency) on A100 GPU(Canopy Labs, n.d.) and further reductions to **25–50 ms** through text

streaming into the model's key–value cache, such performance was not achievable under the available hardware constraints.

For video synthesis, **MuseTalk** required **60–120 seconds** to generate a lip-synced video for the same **13 seconds of audio**, consuming approximately **7 GB of VRAM**. When combined, the end-to-end pipeline incurred **2–3 minutes of latency** to produce a short audiovisual segment. While streaming-based optimizations—such as feeding partial audio chunks into the lip-sync model—were considered, most existing models, including MuseTalk, require **complete audio input before video generation can begin**, making meaningful latency reduction infeasible.

Given the project constraints—particularly **limited computational resources** and a **strict latency budget for real-time conversational interaction**—this approach, despite its high output quality, is unsuitable for real-time deployment and is better suited to **offline generation**. Although aggressive quantization and partial streaming could reduce latency, the fundamental requirement of full-sequence video generation prevents achieving the required **millisecond-level latency**, even with acceptable quality degradation. Consequently, this approach was not pursued further for real-time use under the available hardware conditions.

The second synthesis approach evaluated is **phoneme-driven audiovisual synthesis**. In this pipeline, a lightweight **StyleTTS-based model (Kokoro)** is used to generate speech waveforms while simultaneously predicting **viseme durations**. These viseme timestamps are aligned with the generated speech and passed to **TalkingHead**, which animates mouth movements on a **3D character rendered with Three.js** while playing the audio.

The end-to-end latency of this approach is approximately **1000–1300 ms**. Although this remains relatively high, it is closer to real-time interaction constraints, given that typical human conversational response times are approximately **1500 ms**. The observed latency is largely attributable to executing the models **in-browser using transformer.js**, which cannot fully leverage GPU acceleration compared to a server-side deployment. With further **quantization** and by migrating inference to a **server-based environment**, additional latency reductions are expected. Given the inherent challenges of real-time response generation, the original **1500 ms latency budget** is likely unrealistic. Therefore, the target latency for this project is revised to a more practical range of **2000–3000 ms**. Since inference is performed in the browser, **VRAM usage is not a limiting factor**.

In terms of speech quality, the generated audio is noticeably inferior to that produced by the audio-driven synthesis approach. Despite this reduction in perceptual quality, **latency is prioritized over audio fidelity**, as conversational fluidity is more critical for sustaining interaction. While speech quality remains important—particularly for conveying empathy—the

current trade-off is justified for real-time use. Consequently, additional mechanisms are explored to compensate for the loss in vocal expressiveness.

Two strategies are proposed to address this limitation. First, **emotional expression is not conveyed solely through speech**, but also through visual cues. TalkingHead enables the animation of **facial expressions, head poses, and upper-body gestures**, allowing affective signals to be communicated visually. A limited but expressive set of **six basic emotional states**—*happiness, sadness, anger, fear, disgust, and surprise*—is defined as an initial approximation. These expression labels are provided alongside viseme timings to animate both lip movement and body pose in the 3D character. This multimodal signaling can partially compensate for reduced vocal expressiveness by reinforcing emotional intent through visual channels.

Second, speech expressiveness may be improved by incorporating **style control mechanisms** inspired by Orpheus into the StyleTTS-based pipeline. StyleTTS2, which underlies the Kokoro vocoder, already supports **style vectors derived from reference speech**. By conditioning the model on emotion-specific reference audio (e.g., happy, sad, angry), speech prosody and affect can be modulated while preserving fast generation. These variants will be experimentally evaluated in the later stage of the project to determine the extent to which they reduce the expressiveness gap relative to Orpheus-based synthesis.

Having evaluated the two critical modules, the full system pipeline is assembled using **Zipformer** for speech-to-text, a pretrained **TinyLLaMA** model as the dialogue core, and **Kokoro with TalkingHead** for audiovisual synthesis. This integrated pipeline is designed to achieve an end-to-end latency of approximately **2500 ms**, aligning with the revised real-time interaction constraints identified in earlier evaluations.

In conclusion, this project successfully delivers a functional prototype of a complete multimodal conversational pipeline, incorporating both audio and video processing. A **phoneme-driven synthesis approach** was selected, prioritizing latency over output quality to support real-time interaction. To mitigate the resulting loss in expressiveness, multiple visual and speech-based compensation strategies were proposed. Additionally, the evaluation highlights the necessity of **fine-tuning the dialogue model** to produce concise, conversationally appropriate responses, which remains an important direction for future work.

Model / Module	VRAM Usage	Streaming Support	Total Latency	Time-to-First-Token / Byte
TinyLLaMA	~2 GB	Yes	~2.6 s (258 tokens)	~0.5 s
Orpheus	~6.18 GB	Yes	~51 s (13 s audio)	-
MuseTalk	~7 GB	Limited (requires full audio, but video can stream)	60–120 s (13 s video)	N/A
Kokoro	N/A (browser-based)	Partial	~1000–1300 ms	~300–500 ms
TalkingHead	N/A (browser-based)	Yes	Immediate(render-bound)	Immediate (render-bound)

Table 3.1: Analysis of the models

References

Alessa, A., & Al-Khalifa, H. (2023). Towards Designing a ChatGPT Conversational Companion for Elderly People. *Proceedings of the 16th International Conference on PErvasive Technologies Related to Assistive Environments*, 667–674. https://doi.org/10.1145/3594806.3596572

Balmaceda, J. M., Schiaffino, S., & Godoy, D. (2014). How do personality traits affect communication among users in online social networks? *Online Information Review, 38*(1), 136-153.

Canopy Labs. (2025). *Orpheus-TTS: Towards human-sounding speech* (Version 0.1) [Computer software]. GitHub. https://github.com/canopyai/Orpheus-TTS

Canopy Labs. (n.d.). canopylabs/orpheus-3b-0.1-ft [Computer software]. Hugging Face. https://huggingface.co/canopylabs/orpheus-3b-0.1-ft

Casas, J., Spring, T., Daher, K., Mugellini, E., Khaled, O. A., & Cudré-Mauroux, P. (2021). Enhancing Conversational Agents with Empathic Abilities. *Proceedings of the 21st ACM International Conference on Intelligent Virtual Agents*, 41–47. https://doi.org/10.1145/3472306.3478344

Castelo, N., Boegershausen, J., Hildebrand, C., & Henkel, A. P. (2023). Understanding and improving consumer reactions to service bots. *Journal of Consumer Research, 50*(4), 848-863.

De Freitas, J., Agarwal, S., Schmitt, B., & Haslam, N. (2023b). Psychological factors underlying attitudes toward AI tools. *Nature Human Behaviour, 7*, 1845-1854.

De Freitas, J., Oğuz-Uğuralp, Z., Uğuralp, A. K., & Puntoni, S. (2025). AI Companions Reduce Loneliness. *The Journal of Consumer Research*. https://doi.org/10.1093/jcr/ucaf040.

Dykstra, P. A., Van Tilburg, T. G., & De Jong Gierveld, J. (2005). Changes in older adult loneliness: Results from a seven-year longitudinal study. *Research on Aging, 27*(6), 725-747.

Gravano, A., & Hirschberg, J. (2011). Turn-taking cues in task-oriented dialogue. *Computer Speech & Language*, *25*(3), 601–634. https://doi.org/10.1016/j.csl.2010.10.003

Gulati, A., Qin, J., Chiu, C.-C., Parmar, N., Zhang, Y., Yu, J., Han, W., Wang, S., Zhang, Z., Wu, Y., & Pang, R. (2020). *Conformer: Convolution-augmented Transformer for speech recognition*. arXiv. https://arxiv.org/abs/2005.08100

Hexgrad.Kokoro-82m (revision d8b4fc7), 2025.URL https://huggingface.co/hexgrad/Kokoro-82M.

Holt-Lunstad, J., Robles, T. F., & Sbarra, D. A. (2017). Advancing social connection as a public health priority in the United States. *American Psychologist, 72*(6), 517.

Kwon, W., Li, Z., Zhuang, S., Sheng, Y., Zheng, L., Yu, C. H., Gonzalez, J. E., Zhang, H., & Stoica, I. (2023). Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the ACM SIGOPS 29th Symposium on Operating Systems Principles*. https://doi.org/10.1145/3600006.3613165

K2-FSA. (n.d.). *Sherpa-onnx* (Version 1.12.19) [Computer software]. PyPI. https://pypi.org/project/sherpa-onnx/

Lee, O., & Joseph, K. (2025). *A large-scale analysis of public-facing, community-built chatbots on Character.AI* (arXiv:2505.13354). arXiv. https://arxiv.org/abs/2505.13354

Levinson, S. C., & Torreira, F. (2015). Timing in turn-taking and its implications for processing models of language. *Frontiers in Psychology*, *6*, Article 731. https://doi.org/10.3389/fpsyg.2015.00731

Li, Y. A., Han, C., & Mesgarani, N. (2025). StyleTTS: A Style-Based Generative Model for Natural and Diverse Text-to-Speech Synthesis. *IEEE Journal of Selected Topics in Signal Processing*, *19*(1), 283–296. https://doi.org/10.1109/JSTSP.2025.3530171

met4citizen. (n.d.). *TalkingHead* [Computer software]. GitHub. Retrieved December 16, 2025, from https://github.com/met4citizen/TalkingHead

Pentina, I.; Hancock, T.; and Xie, T. 2023.Exploring Relationship Development with Social Chatbots: A Mixed-Method Study of Replika.*Computers in Human Behavior*, 140: 107600.

Perlman, D., & Peplau, L. A. (1982). Theoretical approaches to loneliness. In L. A. Peplau & D. Perlman (Eds.), *Loneliness: A sourcebook of current theory, research and therapy* (pp. 123-134). Wiley.

Prajwal, K. R., Mukhopadhyay, R., Namboodiri, V. P., & Jawahar, C. V. (2020). A Lip Sync Expert Is All You Need for Speech to Lip Generation In the Wild. *Proceedings of the 28th ACM International Conference on Multimedia*, 484–492. https://doi.org/10.1145/3394171.3413532

Schuller, B. W. (2018). Speech emotion recognition: two decades in a nutshell, benchmarks, and ongoing trends. *Communications of the ACM*, *61*(5), 90–99. https://doi.org/10.1145/3129340

Serengil, S. I. (n.d.). *deepface* [Computer software]. GitHub. Retrieved December 16, 2025, from https://github.com/serengil/deepface

Shahid, A., Chung, J., & Heo, S. (2025). Exploring Older Adults Personality Preferences for LLM-powered Conversational Companions. In N. Yamashita, K. Yatani, X. Ding, & V. Evers (Eds.), *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (pp. 1–7). ACM. https://doi.org/10.1145/3706599.3719976

Skantze, G. (2021). Turn-taking in Conversational Systems and Human-Robot Interaction: A Review. *Computer Speech & Language*, *67*, Article 101178. https://doi.org/10.1016/j.csl.2020.101178

Wang, K. 2024.From ELIZA to ChatGPT: A brief history of chatbots and their evolution.*Applied and Computational Engineering*, 39: 57–62.

Wu, Z., Gong, Z., Ai, L., Shi, P., Donbekci, K., & Hirschberg, J. (2024). *Beyond silent letters: Amplifying LLMs in emotion recognition with vocal nuances*. arXiv. https://arxiv.org/abs/2407.21315

Xie, T.; Pentina, I.; and Hancock, T. 2023.Friend, Mentor, Lover: Does Chatbot Engagement Lead to Psychological Dependence?*Journal of Service Management*, 34(4): 806–828.

Xu, S., Chen, G., Guo, Y.-X., Yang, J., Li, C., Zang, Z., Zhang, Y., Tong, X., & Guo, B. (2024). *VASA-1: Lifelike audio-driven talking faces generated in real time* (arXiv:2404.10667). arXiv. https://arxiv.org/abs/2404.10667

Zhang, W., Cun, X., Wang, X., Zhang, Y., Shen, X., Guo, Y., Shan, Y., & Wang, F. (2023). SadTalker: Learning Realistic 3D Motion Coefficients for Stylized Audio-Driven Single Image Talking Face Animation. *Proceedings (IEEE Computer Society Conference on Computer Vision and Pattern Recognition. Online)*, 8652–8661. https://doi.org/10.1109/CVPR52729.2023.00836

Zhang, Y., Zhong, Z., Liu, M., Chen, Z., Wu, B., Zeng, Y., Zhan, C., He, Y., Huang, J., & Zhou, W. (2025). *MuseTalk: Real-time high-fidelity video dubbing via spatio-temporal sampling*. arXiv. https://arxiv.org/abs/2410.10122

Appendix

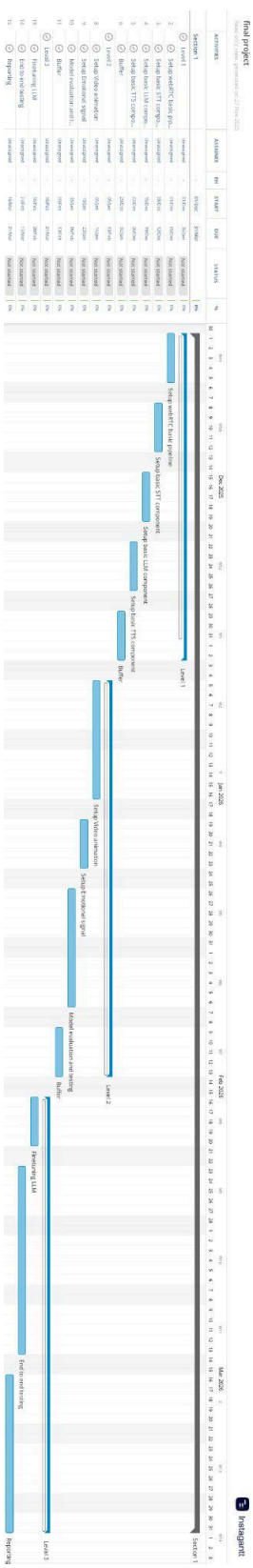
1. **Analysis of existing works on key dimensions**

Dimension	Eldercare voice chatbot	Roleplay social media chatbots	Empathic dialogue systems
Emotional interaction fidelity	Uses ASR → LLM → TTS pipeline. Study reports fluent, coherent responses but mixed engagement; emotional fidelity depends strongly on the unspecified TTS system (some TTS allow expressive cues like sighs/laughter; others do not), so the realized prosodic expressiveness is unclear.	High potential for richly textured interactions because creators craft persona-driven greetings; however, greeting-based design often encodes tropes that can reproduce problematic effects (Lee & Joseph, 2025). Emotional fidelity is uneven and uncontrolled.	Fine-tuned LLMs on empathic dialogues produce high perceived empathy in text-only evaluations (Casas et al., 2021). Fidelity in text is strong, but limited by the absence of vocal/facial cues.
User psychology & context sensitivity	Explicitly targeted at older adults: design choices (voice-first, proactive checks, gamification) align with elder needs; privacy and data-governance issues are salient for this vulnerable group. Evaluation uses human ratings of emotional connection.	Users may form intense parasocial attachments or be exposed to harmful tropes; character creation reflects fandoms and youth-oriented tropes, raising risks for minors and emotionally vulnerable users (Lee & Joseph, 2025).	Models trained on empathic corpora can calibrate emotionally appropriate responses and outperform baselines on perceived empathy, but struggle to maintain multi-turn emotional coherence and to fully model complex user states.
Modality & embodiment requirements	Voice-first modality chosen to reduce keyboard friction for elders (accessibility rationale). Multimodal (audio+video) would likely improve emotion inference, but current study is	Largely text and audio call based; some characters simulate embodiment via descriptive cues. Role-play emphasis substitutes persona design for true multimodal	Mostly text-based (Casas et al., 2021) or chat UI with possible voice/TTS extensions in applied settings. Text limits perceptual access to tone, hesitation, micro-expressions—a

	voice-only; unspecified TTS affects embodiment.	embodiment, which can amplify problematic content.	reas where audio/video would add value.
Personalization & identity construction	Personalization via onboarding questionnaire. Voice and role selection are important but must be handled with privacy safeguards for elders.	Extremely flexible personalization—user-created characters enable many identities and tropes; lack of moderation can produce identities that reinforce harmful norms.	Personalization via fine-tuning or persona conditioning is effective for empathetic behavior, but over-personalization risks dependency or misaligned expectations in commercial contexts.
Safety, ethics & attachment risks	High risk/ethical sensitivity: elders are vulnerable; must ensure safe responses, data governance, and limits on dependency. Paper cautions about ChatGPT's limits and calls for safety testing.	High risk of amplifying toxic content, gendered violence tropes, and unqualified therapeutic claims (Lee & Joseph, 2025). Can normalize harmful dynamics and mislead minors.	Risks include inappropriate or misleading empathic responses if the model hallucinates or overconfidently assumes therapeutic authority. Customer-facing bots risk negative brand perception if perceived as prioritizing firm efficiency (Castelo et al., 2023; De Freitas et al., 2023a).
Evaluation & metrics	Human rating scales for fluency, engagement, and interest; shows high fluency but weaker engagement—suggests need for more nuanced metrics (longitudinal well-being, safety).	Analysis based on large-scale scraping of character bios and greetings (Lee & Joseph, 2025); lacks direct user interaction metrics but reveals cultural/trope signals that should inform moderation.	Human judgments of empathy (Casas et al., 2021) show gains vs. baselines; also measurable business/KPI impacts in customer settings (Bergner et al., 2023), but multi-turn empathy consistency remains an open evaluation challenge.
Deployment / practical considerations	Scalable and accessible (voice + phone); however, production requires	Highly scalable user-generated ecosystem but low moderation overhead	Feasible to integrate into customer support and mental-health adjuncts; requires

	robust ASR in noisy/older-adult contexts, tightly controlled TTS, and strict privacy controls for sensitive personal data.	creates safety/legal exposures; content moderation and age gating are practical challenges.	domain-specific fine-tuning and safety layers to avoid hallucinations and harmful advice.

1. Gantt Chart of work plan



1. **Pretrained LLM Response**

```
<|system|>
# Speak in short sentences.</s>
<|user|>
# What is the meaning of life?</s>
<|assistant|>
# The meaning of life is not a specific meaning or concept, but a question
which is constantly being posed and discussed by philosophers and
scientists alike.

    # Philosophers have long debated the meaning of life, seeking to find
answers to questions like:
    - What is the purpose of existence?
    - Is there a higher power or being that we can seek guidance from?
    - Do we have free will or are we destined to follow predetermined
paths?

    # Scientists have also explored the question of the meaning of life,
from studying the origins and evolution of life on earth to studying the
ways in which we interact with the universe
    # The meaning of life is not a specific meaning or concept, but a
question which is constantly being posed and discussed by philosophers and
scientists alike.

    # Philosophers have long debated the meaning of life, seeking to find
answers to questions like:
    - What is the purpose of existence?
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    - Do we have free will or are we destined to follow predetermined
paths?

    # Scientists have also explored the question of the meaning of life,
from studying the origins and evolution of life on earth to studying the
ways in which we interact with the universe.
```